Multi-Dimensional Array in Java

03 November 2026 20:14

Multidimensional array in Java are used to store data in the form of rows and columns.

What are multi - dimensional arrays in Java?
Multi-dimensional array or arrays of arrays where each element of the main array holds another array. They are commonly used to represent data in table or matrix form. Such as grids, charts or spreadsheets.

A multi-dimensional array can be compared to a table. For example, storing marks of students where each row represents a student and each column represents a subject. This way data is organized in rows and columns using a multi-dimensional array.

Two-dimensional (2D) Array

A two dimensional array (2D) array is a type of multi-dimensional array that stores data in the form of rows and columns. It is used to represent table - like structures such as matrices, grids or tables where data is organized in two dimensions only.

Declaration
You can use one of the following syntaxes to declare a 2-dimensional array.

```
DataType[][] arrayName;

Or

DataType arrayName[][];

Allocating memory:

arrayName = new DataType[rows][columns];
```

We can declare and allocate memory for two dimensional array in a single statement as follows:

```
Datatype[][] arrayName = new DataType[rows][columns];

            Or
Datatype arrayName[][] = new DataType[rows][columns];
```

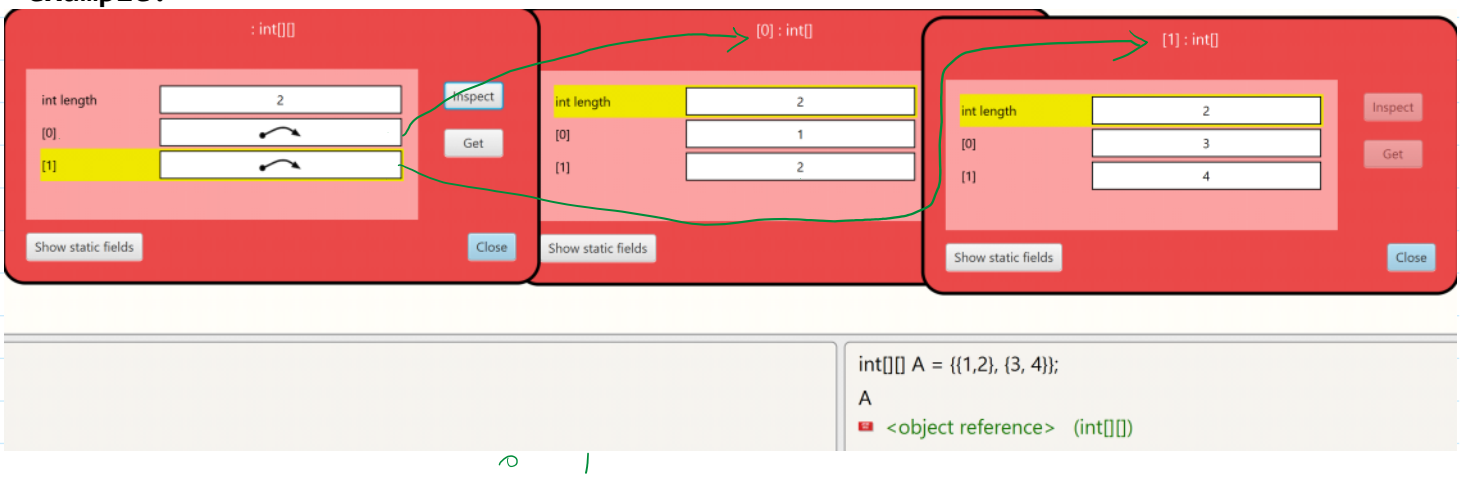
Assigning values (initialization)

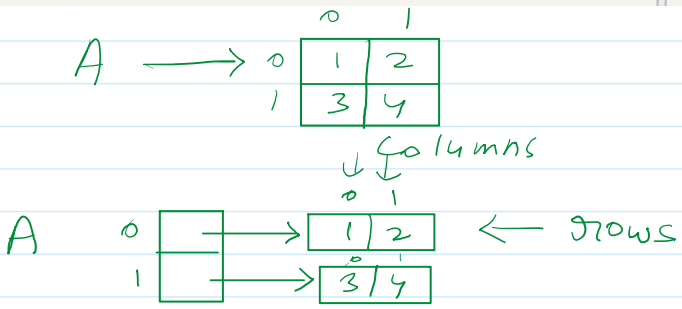
```
Datatype arrayName[][] = {{v1, v2, v3…},{…},…};
```

For example:



For example:





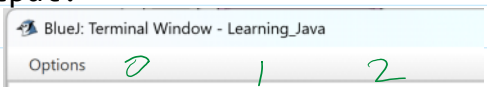

```
package MyArray.MultiDimension;


/**
 * Write a description of class One here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class One
{
    public static void main(String[] args){
        System.out.print("\f");
        int[][] A = {{1,2,3}, {4, 5, 6}};
        int i, j;
        for(i = 0; i < A.length; i++){

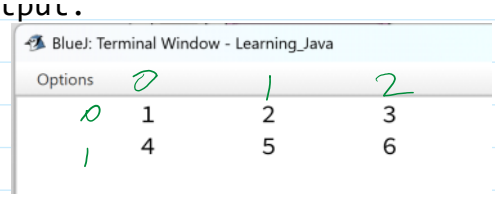
            for(j = 0; j < A[i].length; j++){
                System.out.print("\t" + A[i][j]);
            }
            System.out.print("\n");
        }
    }
}
```

number of rows
number of columns
or
number of elements in a row
arrayName.length
arrayName[i].length

Output:



Output:




```
package MyArray.MultiDimension;
import java.util.*;

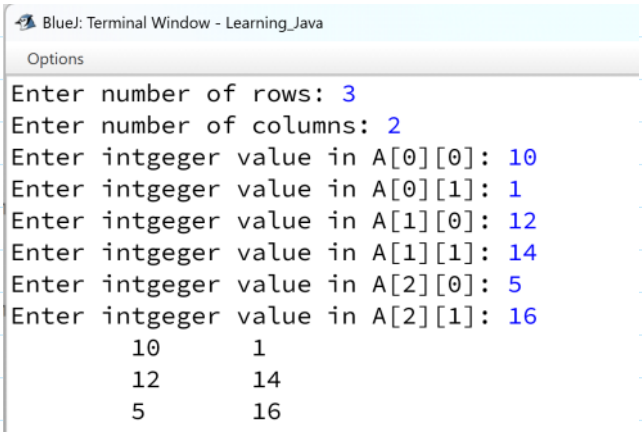

/**
 * Write a description of class UserInput here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class UserInput
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("\fEnter number of rows: ");
        int rows = sc.nextInt();
        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();

        int[][] A = new int[rows][cols];
        int r, c;
        for(r = 0; r < A.length; r++){
            for(c = 0; c < A[r].length; c++){
                System.out.print("Enter intgeger value in A["+r+"]["+c+"]:
");
                A[r][c] = sc.nextInt();
            }
        }

        //Display 2-D array
        for(r = 0; r < A.length; r++){
            for(c = 0; c < A[r].length; c++){
                System.out.print("\t" + A[r][c]);
            }
            System.out.print("\n");
        }

    }
}
```

Output:



WAP to create a 2-D Array by taking number of rows and number of columns, and print the 2-D array. Find the sum of each row and each column.

1	2	3	= 6
4	5	6	= 15
5	7	9	